Prompt Sensitivity and Silent Action Drift in OpenVLA: A Two-Stage Study on BridgeData V2

HARMIT PATEL

Abstract

Vision-language-action (VLA) models promise a simple interface for robot control: the model sees an image, reads a language instruction, and predicts the next robot action. Recent work, however, suggests that current VLA systems do not always use language as reliably as their task success numbers imply. This paper studies that problem in OpenVLA-7B with a fully reproducible local evaluation pipeline. We run two complementary experiments on BridgeData V2 images. Manual40 contains 40 hand-selected trajectories with manually written task prompts. Auto50 contains 50 automatically sampled trajectories with prompts derived from run-family metadata. For every image we evaluate four prompt families: normal, paraphrased, contradictory, and neutral. We compare the predicted actions using the first six action dimensions, measuring both L_2 drift and cosine similarity relative to the normal prompt. Across both experiments, paraphrases preserve behavior best, while contradictory and neutral prompts induce larger drift. Final-state images are more fragile than initial-state images in both branches. Manual40 shows mean 6D L_2 drift of 0.028/0.042/0.045 on initial frames and 0.043/0.060/0.071 on final frames for paraphrased/contradictory/neutral prompts. Auto50 shows higher overall drift at 0.056/0.070/0.073 on initial frames and 0.081/0.094/0.082 on final frames. The paper does not claim a full safety solution. Instead, it offers a careful empirical slice of prompt robustness, a working local pipeline, and a concrete argument for using action drift as a future inference-time safety signal.

1 Introduction

Large vision-language-action models promise a simple idea: show the robot a camera image, tell it what to do in words, and let it predict the next robot action. OpenVLA is one of the most visible open-source examples of this idea and has become attractive because it can be run, adapted, and inspected by researchers without needing a closed industrial stack [1]. At the same time, recent work has started to question how much these models truly depend on language once the scene is visually informative. Benchmarks such as LIBERO-X show that robustness can drop sharply under structured perturbations [6]. LangGap argues that current VLAs often ignore language entirely unless the benchmark forces language understanding by design [7]. IGAR shows that even contradictory instructions may still produce visually plausible actions, a failure mode it calls linguistic blindness [8]. LangForce explains part of this behavior as information collapse, where the instruction becomes too predictable from the image and therefore stops influencing the action strongly enough [9].

This paper studies that problem from a practical, reproducible perspective. Instead of building a new benchmark from scratch, we take an existing open model, OpenVLA-7B, and test its sensitivity to four prompt families on BridgeData V2 images [4]. We run two complementary evaluation branches. The first branch, which we call Manual40, contains 40 hand-selected trajectories with task-specific language written manually from the scene. The second branch, Auto50, contains 50 automatically sampled trajectories from the scripted BridgeData split and uses language prompts derived from the run-family metadata. For each image

we compare a normal instruction against a paraphrased version, a contradictory version, and a neutral version. We then measure action drift relative to the normal prompt using the first six action dimensions, and we compare initial and final states separately.

Our contribution is modest but concrete. First, we provide a local, fully reproducible OpenVLA evaluation pipeline that runs under WSL with 4-bit loading and a patch for the image-text attention-mask mismatch encountered during inference. Second, we report a two-stage prompt-sensitivity study over 180 image evaluations in total: 80 images from Manual40 and 100 images from Auto50. Third, we show a consistent pattern across both branches: paraphrases preserve behavior best, contradictory and neutral prompts induce larger drift, and final-state images are more fragile than initial-state images. We do not claim to solve language robustness in VLAs. Instead, we provide a careful empirical slice of the problem, a concrete engineering recipe for reproducing the study, and a clear agenda for the next step: turning action drift into an inference-time safety signal.

2 Related Work

OpenVLA is the core model family for this study because it is an open 7B VLA trained on large-scale robot demonstrations and designed to be fine-tunable on consumer hardware [1]. More broadly, open robot-learning infrastructure is maturing quickly; LeRobot is a recent example of a vertically integrated open-source stack that connects data, robots, policies, and deployment tooling [2]. The larger VLA literature is also expanding rapidly, and a recent survey organizes the field around multimodal integration, tokenization, learning paradigms, applications, and open challenges [3]. Our work focuses on one specific open challenge inside that broader landscape: language robustness at inference time.

Several recent papers are especially close to our problem setting. LIBERO-X argues that existing VLA benchmarks can hide weaknesses in instruction grounding and proposes a robustness benchmark with progressive difficulty [6]. LangGap shows that state-of-the-art VLAs can largely ignore language unless the benchmark forces multiple semantic tasks within the same layout [7]. IGAR studies contradictory instructions and proposes train-free attention recalibration to restore instruction influence [8]. LangForce frames the issue as information collapse, where instructions become too predictable from the image and therefore stop contributing enough mutual information to the action policy [9]. SAFE and FailSafe move the discussion toward runtime safety by focusing on failure detection and recovery [10, 11]. Q-DIG red-teams VLA prompts by generating diverse failure-inducing instructions in simulation [12]. Relative to this literature, our contribution is narrower but concrete: a reproducible two-stage OpenVLA study that compares paraphrase, contradiction, and neutral prompting across initial and final states, using both a manual and an automatically sampled BridgeData branch.

3 Research Questions

We study four questions.

1. Does OpenVLA preserve its predicted action under meaning-preserving paraphrases?
2. How much larger is the drift under contradictory or underspecified prompts?
3. Are final-state images more fragile than initial-state images?

4. Does the main pattern survive when moving from a hand-curated subset to a broader automatically sampled subset?

4 Experimental Setup

extbfModel and runtime. All experiments use OpenVLA-7B in 4-bit mode on a local WSL environment backed by an NVIDIA RTX 2060 GPU. The base model is loaded through the Hugging Face OpenVLA checkpoint with trust-remote-code enabled. We use the standard processor and the model's built-in extttpredict_action interface.

extbfInference patch. During initial testing we hit a critical mismatch between the text attention mask and the image-augmented embedding sequence. The practical symptom was a runtime failure after the visual tokens were fused into the transformer input. The fix was to prepend a 256-token image attention mask before calling extttpredict_action. This patch does not change the OpenVLA weights; it only repairs the alignment expected by the model's inference path. This engineering step matters because without it the study would not be reproducible on the target local hardware.

extbfData. The first evaluation branch, Manual40, was built manually from 40 BridgeData V2 trajectories. For each trajectory we saved one initial frame and one final frame, resulting in 80 total images. We then wrote one task-specific natural-language command per trajectory and derived three additional prompt variants from it. The second branch, Auto50, was created automatically from the downloaded scripted BridgeData split. We sampled 50 trajectories, extracted one initial frame and one final frame for each, and generated task prompts from the recorded run-family metadata. This produced 100 total images. The combined study therefore covers 180 image evaluations.

extbfPrompt families. Each image is evaluated with four prompt types: (1) *normal*, which states the intended task directly; (2) *paraphrased*, which keeps the meaning but changes the wording; (3) *contradictory*, which intentionally conflicts with the original task; and (4) *neutral*, which removes task-specific information and asks only what action the robot should take. The central question is whether the action predicted by OpenVLA remains stable under meaning-preserving changes and how strongly it shifts under meaning-breaking or underspecified prompts.

extbfMetrics. We measure drift using only the first six action dimensions, excluding the final scalar that often collapsed to a near-binary value in some runs. For each non-normal prompt, we compute (a) the L_2 distance from the normal action and (b) the cosine similarity with the normal action. Lower L_2 means less drift; higher cosine means more directional agreement. We also compute simple robustness counts: exact-all-same cases, near-all-same cases with max drift below 0.01, the number of times the paraphrase is the closest prompt to the normal action, and the number of times contradictory or neutral prompts are the farthest.

extbfAnalysis outputs. We summarize the means, 95% bootstrap confidence intervals, and a ranked list of high-drift cases. We deliberately keep the study simple: no simulator rollouts, no success-rate labels, and no active safety intervention. This makes the present paper a diagnosis-first study rather than a complete safety system.

5 Results

Table 1 gives the main quantitative summary. In all four groups, paraphrase drift is the smallest. Final-state images are more fragile than initial-state images in both Manual40 and Auto50.

Table 1: Main summary across both experiments. Lower L_2 is better; higher cosine is better.

Exp.	Group	n	L2-P	L2-C	L2-N	Cos-P	Cos-C	Cos-N
Manual40	Initial	40	0.028	0.042	0.045	0.679	0.586	0.426
Manual40	Final	40	0.043	0.060	0.071	0.480	0.219	-0.002
Auto50	Initial	50	0.056	0.070	0.073	0.484	0.269	0.171
Auto50	Final	50	0.081	0.094	0.082	0.387	0.157	0.220

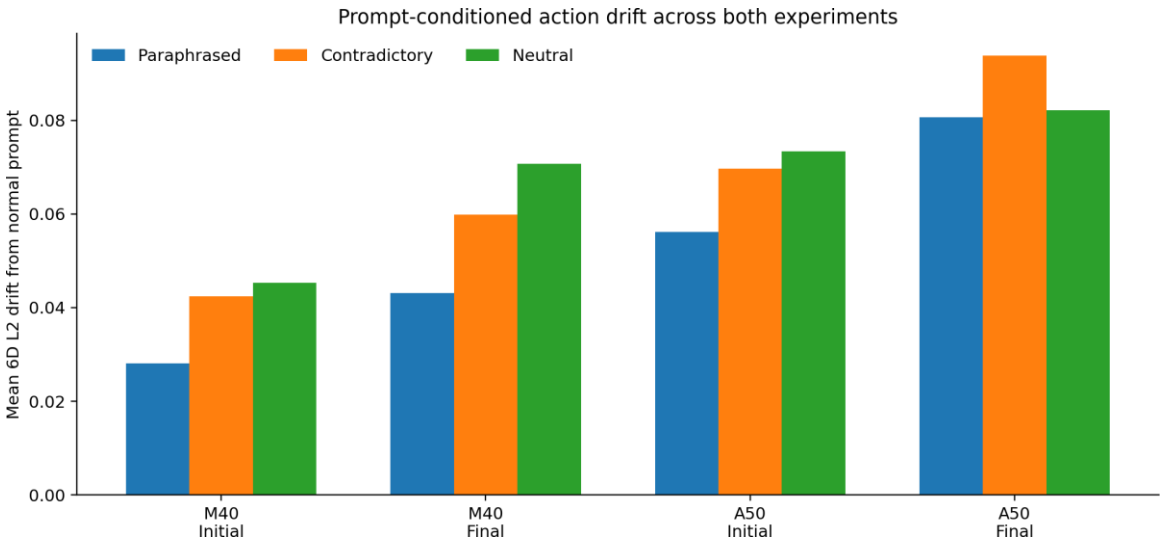

Figure 1: Mean 6D L_2 drift from the normal prompt. Paraphrases are always closest. Final scenes are always more fragile than initial scenes.

6 Discussion

The results are consistent enough to support three claims. First, paraphrases are the most stable prompt family in every group we studied. Second, contradictory and neutral prompts create materially larger drift than paraphrases, though the ordering between contradictory and neutral depends on the branch and state. Third, final-state images are more fragile than initial-state images in both Manual40 and Auto50.

In Manual40, the average 6D L_2 drift rises from 0.028 under paraphrases on initial frames to 0.043 on final frames. Contradictory drift rises from 0.042 to 0.060, and neutral drift rises from 0.045 to 0.071. This branch therefore shows a clean monotonic fragility pattern from initial to final states. The same story largely repeats in Auto50 but at a higher overall drift level. Auto50 initial-frame drift is 0.056 for paraphrases, 0.070 for contradictions, and 0.073 for neutral prompts. Auto50 final-frame drift increases further to 0.081 for paraphrases, 0.094 for contradictions, and 0.082 for neutral prompts. The final-state increase is therefore not confined to the manual branch; it survives the shift to a broader automatically sampled dataset.

The cosine results tell the same story from a directional rather than magnitude-based perspective. In Manual40 initial frames, paraphrase cosine remains high at 0.679, while contradictory and neutral drop to 0.586 and 0.426. In Manual40 final frames the drop becomes sharper, especially for contradictory and neutral prompts, with the neutral mean crossing slightly below zero. Auto50 cosine values are lower overall, again suggesting that the broader sampled branch is harder. Yet even there, paraphrases preserve the strongest alignment in both initial and final groups.

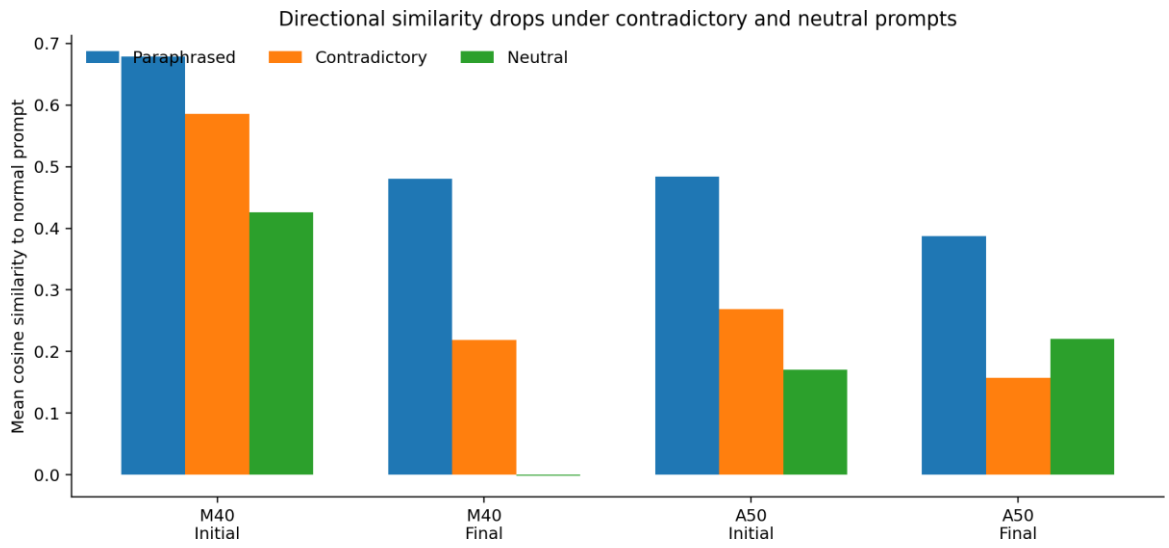

Figure 2: Mean cosine similarity to the normal prompt. Directional agreement decreases most strongly under contradictory and neutral prompt changes.

Table 2: Manual40 robustness counts. Near-all-same means max drift < 0.01.

Group	Exact-all-same	Near-all-same	Para closest	Contra farthest	Neutral farthest
Initial	1	8	25	14	21
Final	0	4	22	12	21

The invariance counts add an important nuance. In Manual40 we still observe one exact-all-same case and several near-all-same cases, especially on initial frames. Auto50 almost eliminates this effect, with only one near-all-same case in initial frames and one in final frames. This suggests that the manual subset contains some images where the task is visually overdetermined, but once the sampled branch becomes broader, OpenVLA is less likely to collapse to identical outputs across all prompts. In other words, the model is not uniformly blind to language; it is inconsistently sensitive, which may be even more dangerous in practice because the failure mode is scene-dependent rather than constant.

Finally, the top-failure ranking shows that the largest drifts concentrate in final scenes and in pick-and-place families involving soft toys, bins, and crowded object layouts. That pattern is intuitive. Final scenes often look more ambiguous because the goal may already be partly achieved, objects may be occluded, and the action space may contain both completion and correction motions. Under these conditions, a language change can more easily flip the model from a stable continuation policy to a qualitatively different adjustment policy. This observation points toward a useful next step: drift itself may be an early warning signal for prompt confusion.

7 Limitations and Future Work

This study has four important limitations. First, it evaluates action-space drift rather than true task success. We can show that the predicted action changes, but we cannot yet say how often that change would produce an execution failure, a recoverable deviation, or a harmless alternative trajectory. A simulator study or

Table 3: Auto50 robustness counts. Near-all-same means max drift < 0.01.

Group	Exact-all-same	Near-all-same	Para closest	Contra farthest	Neutral farthest
Initial	0	1	27	15	22
Final	0	1	26	16	15

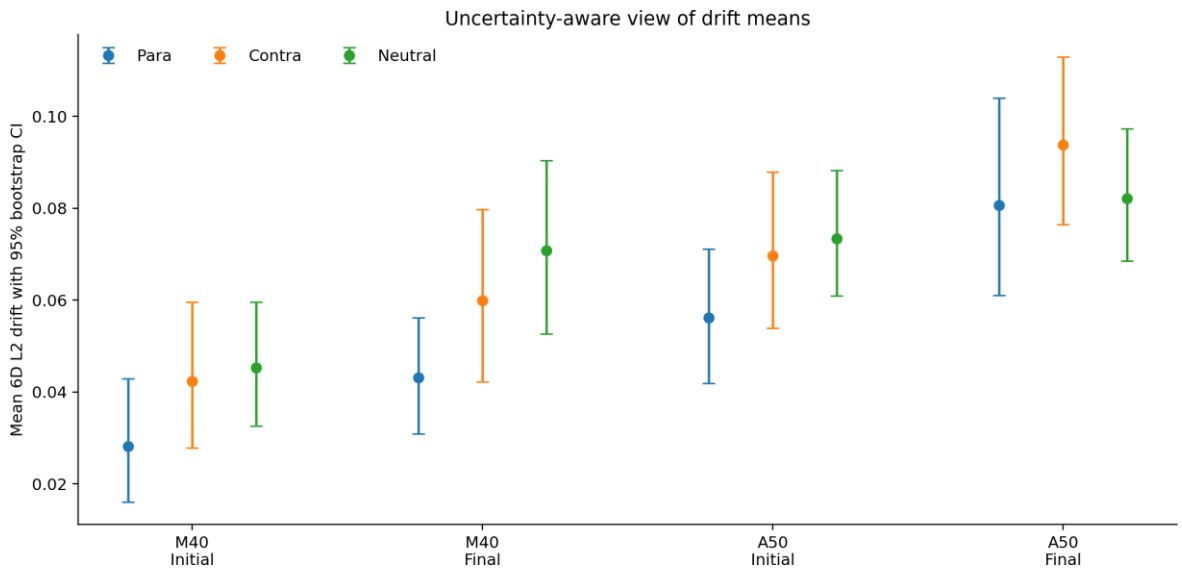

Figure 3: Mean 6D L_2 drift with 95% bootstrap confidence intervals. The intervals overlap less between paraphrased and the more disruptive prompt families in the harder groups, especially Manual40-Final and Auto50-Final.

real-robot replay is needed for that step. Second, we evaluate one base model, OpenVLA-7B, rather than a cross-model panel. This makes the paper cleaner and more reproducible, but it limits claims about whether the same pattern holds across other VLAs such as π_0-FAST or OpenVLA-OFT. Third, the Manual40 and Auto50 branches are not identical in construction: one uses task-specific manual prompts, while the other derives language from run-family metadata. This is a strength for breadth but a limitation for strict apples-to-apples comparison. Fourth, we patched an inference-time attention-mask issue to make the model run locally. Although this does not alter the learned weights, any local engineering patch raises the question of whether hidden implementation details could affect robustness.

These limitations also explain why the problem is not already solved. Prompt robustness in VLAs is hard for at least three reasons. The first is data bias: in many robot datasets the scene already predicts the task strongly enough that the model can succeed by following visual shortcuts, a problem emphasized by LangForce and LangGap [7, 9]. The second is evaluation cost: measuring language sensitivity with real robots or large simulation suites is expensive, and fine-grained language perturbations multiply that cost quickly. The third is systems complexity: once a model mixes images, text, and low-level control, failures can originate from the data, the fusion architecture, the action head, or the runtime implementation itself.

Future work should therefore move in three directions. The first is *outcome-aware evaluation*: run the highest-drift cases in simulation and test whether drift predicts task failure. The second is *inference-time safety*: treat drift as a detector that can trigger re-prompting, abstention, or a safe retreat behavior, much like the failure-detection literature represented by SAFE and FailSafe [10, 11]. The third is *stronger causal*

Table 4: Selected high-drift cases ranked by maximum drift.

Experiment	Group	Image	Task	L2-P	L2-C	L2-N
Auto50	Final	traj015_final	pick up the soft toy	0.362	0.166	0.189
Auto50	Final	traj004_final	place object in bin	0.297	0.253	0.100
Manual40	Final	traj012_final	pick up the paper	0.092	0.239	0.294
Auto50	Initial	traj024_initial	pick up the object	0.144	0.285	0.225
Auto50	Final	traj049_final	pick up the utensil	0.034	0.278	0.048
Manual40	Initial	traj024_initial	move can right	0.215	0.167	0.067
Manual40	Final	traj019_final	put green object into drawer	0.068	0.240	0.073
Manual40	Initial	traj031_initial	open brown box flap	0.077	0.201	0.073

testing: extend the study across multiple VLA families and across structured benchmarks such as LIBERO-X, which were designed to expose language and scene-understanding weaknesses more systematically [6].

8 Conclusion

We presented a fully reproducible prompt-sensitivity study for OpenVLA-7B using two BridgeData V2-based evaluation branches. Manual40 provides a hand-curated, task-specific view; Auto50 provides a broader sampled validation view. Across both branches, paraphrases preserve action behavior best, while contradictory and neutral prompts produce larger drift. Final-state images are consistently more fragile than initial-state images. The results do not prove that OpenVLA ignores language everywhere, but they do show that language sensitivity is uneven, scene-dependent, and weak enough to matter. That is already useful: a robot policy does not need to fail on every contradictory prompt to be unsafe. It only needs to fail unpredictably. Our study offers a concrete local pipeline, a set of reproducible results, and a practical basis for the next step: turning action drift into a real-time warning signal for safer VLA deployment.

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A Implementation Notes

The local pipeline is important for reproducibility because the engineering details were non-trivial. The model was executed under WSL with CUDA-enabled PyTorch and 4-bit loading. The critical patch prepended a 256-token image mask before inference. Manual40 prompts were written trajectory by trajectory. Auto50 prompts were generated from the scripted metadata and then expanded into initial and final image rows. Analysis scripts computed summary tables, bootstrap confidence intervals, and ranked failure cases. The final package includes the paper source, figures, summary CSVs, and the key result files so that the study can be re-opened later without reconstructing the full debugging history from scratch.

B Recommended Next Step for a Stronger Submission

If this work is extended for a higher-risk conference submission, the highest-value next step is not more prompt variations but an outcome-aware safety layer. The most direct upgrade would be to evaluate the highest-drift cases in simulation and test whether drift predicts task failure. A second upgrade would use drift as a runtime detector that triggers a safer action such as re-prompting, abstaining, or retreating. Those additions would move the paper from diagnosis alone toward diagnosis plus intervention, which is better aligned with recent safety-oriented VLA work.